

- 4 In 1973, a technique for genetic engineering was used for the first time. Recombinant DNA was made using a plasmid and this was successfully transferred into an organism.

In 2012, a new technique for genetic engineering, called gene editing, was developed.

- (a) Table 4.1 lists some statements about the two genetic engineering techniques.

Complete Table 4.1 to compare the original genetic engineering technique using a plasmid vector with the newer technique of gene editing.

For each row, place a tick (✓) in the correct column if the statement applies and leave a blank if the statement does not apply.

Table 4.1

statement	genetic engineering using a plasmid	gene editing
It can add a new phenotypic characteristic to an organism.		
It can change an A–T base pair to C–G.		
It can inactivate a desired selected gene in an organism.		
It may change DNA in a way that cannot be distinguished from a natural mutation.		
It requires a DNA donor and a recipient.		

(b) *Camelina sativa* is a fast-growing plant with oil-rich seeds.

C. sativa grows in dry and poor soils and so it may be important as a food crop in the future. The oil from its seeds has a high content of polyunsaturated fatty acids. This shortens the time that the oil can be stored for, which is a disadvantage.

Scientists used gene editing to develop two types of *C. sativa* with different genetic changes. The gene edited *C. sativa* seeds produced oil with longer storage times.

Fig. 4.1 shows the percentage composition of fatty acids in the oil extracted from seeds of gene edited and wild type (not gene edited) *C. sativa*.

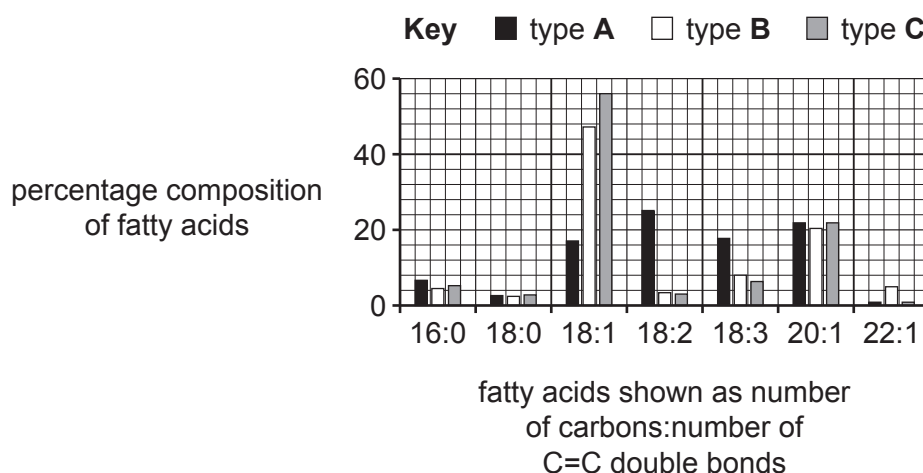


Fig. 4.1

(i) Identify the letter that represents the oil of the wild type *C. sativa* on Fig. 4.1.

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(ii) With reference to Fig. 4.1, discuss the social benefits of this example of gene editing.

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[Total: 9]